

REVIEW ARTICLE

Archaeology as an Integrative Science of the Human Past: Methods, Evidence, Ethics, and Future Directions

SAMUELSON G

Independent Researcher

Corresponding author: SAMUELSON G

Email: gsamuelsonguna@gmail.com

ORCID: 0009-0005-8744-8178

Manuscript type	Interdisciplinary narrative review
Running title	Integrative Archaeology
Literature coverage	Landmark and contemporary sources available through July 2026

Abstract

Archaeology reconstructs human lives, institutions, technologies, and environments from material traces whose meaning depends on context. The discipline is therefore neither simply excavation nor the collection of old objects: it is an inferential science and humanistic practice that integrates stratigraphy, landscape survey, chronology, materials characterization, bioarchaeology, environmental evidence, documentary sources, computation, and community knowledge. This review synthesizes the conceptual foundations and principal methods of contemporary archaeology, with attention to how evidence is generated, transformed, and interpreted. It explains the central importance of provenience, association, formation processes, sampling, uncertainty, and multiple working hypotheses; compares major survey, prospection, excavation, dating, and laboratory techniques; and evaluates the opportunities and limits of LiDAR, geophysics, photogrammetry, ancient DNA, stable isotopes, proteomics, and artificial intelligence. Because archaeological investigation is often destructive and frequently concerns living communities, ancestors, contested heritage, and illicit markets, the review treats ethics and governance as methodological requirements rather than administrative additions. It incorporates stewardship, community partnership, legal compliance, minimal intervention, transparent uncertainty, FAIR data, long-term curation, and accountable communication into a unified research lifecycle. The paper concludes that the strongest archaeological explanations emerge from convergent, independently testable evidence linked to explicit models of site formation and human action. Future progress depends less on any single technology than on responsible integration: pairing high-resolution instruments with well-designed questions, preserving contextual information, involving rights-holders from project inception, and making claims reproducible without exposing sensitive sites or culturally restricted knowledge.

Keywords: archaeology; archaeological method; stratigraphy; archaeological science; remote sensing; radiocarbon dating; ancient DNA; digital archaeology; heritage ethics; community archaeology

Highlights

- Archaeological knowledge depends on context, formation processes, sampling design, and explicit uncertainty—not on artifacts in isolation.
- Survey, excavation, dating, biomolecular science, and digital methods are complementary and should be selected by question, scale, preservation, and ethical cost.
- Stewardship, community authority, legal provenance, and long-term data curation are integral to valid archaeological method.

- The most defensible interpretations use convergent evidence, alternative hypotheses, and reproducible analytical workflows.

1. Introduction

Archaeology is the study of the ancient and recent human past through material remains. Its evidence includes artifacts, architecture, sediments, human and animal remains, plant residues, modified landscapes, industrial debris, inscriptions, and digital records. The temporal range extends from the earliest traces of hominin behavior to the material culture of the contemporary world; consequently, archaeology is defined less by age than by its engagement with the material consequences of human action (Society for American Archaeology, n.d.; Renfrew & Bahn, 2020).

The discipline occupies an unusual epistemic position. Archaeologists investigate events and practices that cannot be directly repeated, while the surviving record is partial, spatially uneven, and altered by biological, geological, chemical, and social processes. Excavation can clarify these processes, but it also removes deposits in an irreversible sequence. Archaeological method must therefore convert destructive observation into durable records, samples, collections, and arguments. The quality of an interpretation depends on the chain connecting a question to field design, contextual recording, laboratory procedures, chronological modelling, comparative reasoning, and public stewardship.

Archaeology has changed substantially from antiquarian traditions focused on spectacular objects. Culture-historical approaches organized material variation in time and space; processual archaeology emphasized explanation, systems, ecology, and hypothesis testing; post-processual and interpretive approaches highlighted meaning, agency, identity, power, and the situated character of knowledge; Indigenous, feminist, Black, community-based, and decolonial archaeologies challenged extractive authority and broadened who defines research questions and owns the outcomes (Binford, 1962; Hodder, 1999; Smith & Wobst, 2005; Trigger, 2006). Contemporary practice draws selectively from these traditions rather than following a single universal theory.

At the same time, archaeological science has expanded the observable record. Radiocarbon calibration and Bayesian chronology can resolve sequences probabilistically; stable isotopes can inform diet, mobility, and environmental conditions; microscopy and spectroscopy characterize production and exchange; ancient DNA and proteins reveal biological histories; and remote sensing can map archaeological landscapes at scales impossible through pedestrian survey alone (Bronk Ramsey, 2009; Makarewicz & Sealy, 2015; Orlando et al., 2021; Reimer et al., 2020; Vinci et al., 2025). Yet technological

power can amplify poor sampling, weak provenance, inequity, or overconfident inference. Methodological integration is therefore both a scientific and ethical problem.

This paper provides a field-wide review organized around four questions: What constitutes archaeological evidence? How is that evidence discovered, dated, analyzed, and interpreted? What limits the strength of archaeological claims? And what responsibilities arise because archaeological records are finite, frequently contested, and connected to living people? The aim is not to replace specialist manuals but to provide a coherent framework suitable for research design, teaching, interdisciplinary collaboration, and manuscript preparation.

2. Review Design and Scope

This is a structured narrative review, not a systematic review or meta-analysis. Sources were identified through targeted searching of scholarly publisher platforms, professional archaeological organizations, heritage bodies, and research-data repositories, supplemented by citation chaining from established methodological works. Priority was given to landmark theoretical and field-method texts, peer-reviewed primers and reviews, major dating and analytical references, and current official guidance on ethics, illicit trafficking, community relations, and data stewardship. The literature cut-off for web-accessible guidance was July 2026.

The review is global in orientation but cannot represent every regional legal system, intellectual tradition, or community protocol. Archaeology is practiced under different heritage laws, land-tenure arrangements, museum histories, and relationships between researchers and descendant communities. Accordingly, the paper emphasizes transferable principles while repeatedly requiring local legal and cultural authority. Technical performance values are presented qualitatively because detection depth, precision, sample requirements, and cost vary with instrument, material, sediment, climate, operator experience, and research design.

No archaeological site was excavated, no human or animal material was sampled, and no original dataset was generated for this manuscript. Figures are original conceptual syntheses created for the review; they are not empirical measurements. Tables summarize method selection and are intended as decision aids rather than fixed rankings.

Table 1. Major classes of archaeological evidence and their interpretive vulnerabilities.

Evidence class	Examples	Primary questions	Key contextual risks
Artifacts	Ceramics, lithics, metals, glass, tools, ornaments	Technology, chronology, exchange, use, identity	Residuality, redeposition, collection bias, typological circularity
Features and architecture	Hearths, pits, walls, roads, fields, drains	Activities, organization, construction sequence, settlement form	Truncation, reuse, repair, later disturbance
Ecofacts and sediments	Seeds, charcoal, pollen, phytoliths, fauna, soils	Diet, economy, seasonality, climate, land use	Intrusion, differential preservation, recovery method
Human remains	Skeletons, teeth, cremations, mummified tissue	Health, demography, mobility, diet, kinship	Ethics, destructive sampling, curation history, population inference
Landscapes	Terraces, canals, routeways, quarries, field systems	Human–environment interaction, connectivity, territoriality	Scale mismatch, visibility bias, modern land-use masking
Documents and inscriptions	Texts, maps, photographs, archives, oral histories	Names, institutions, chronology, emic categories	Authorship bias, preservation, translation, colonial archives
Digital traces	Born-digital records, remote-sensing data, 3D models	Recent materiality, reproducibility, monitoring	Obsolescence, proprietary formats, privacy, sensitive coordinates

3. Archaeological Evidence and Inference

3.1 Context, provenience, and association

An object becomes archaeological evidence through its relationships. Provenience specifies where an item was found, normally in three dimensions and within a defined recording system. Context includes the depositional unit, feature, stratigraphic relationships, matrix, assemblage, and recovery history. Association concerns whether

materials genuinely belonged to the same event or behavioral system. A bronze object without secure provenience may still be chemically or stylistically interesting, but it loses much of its capacity to date an activity, reconstruct a household, or test a behavioral hypothesis.

The basic unit of excavation is therefore not the object but the context. Context-based excavation records deposits, cuts, structures, fills, surfaces, interfaces, and samples as distinct entities. Their physical relationships can be represented in a stratigraphic matrix, which separates sequence from interpretation and permits later revision. This practice operationalizes the law of superposition while accommodating cuts, inversions, redeposition, and complex architecture (Harris, 1989).

3.2 Formation processes and equifinality

The archaeological record is not a direct photograph of past behavior. Cultural formation processes include discard, abandonment, curation, recycling, rebuilding, looting, and excavation itself. Non-cultural processes include erosion, sedimentation, root action, burrowing, chemical dissolution, freeze–thaw cycles, fire, water transport, and differential preservation. Formation-process analysis asks how materials entered the record, what happened after deposition, and which observations are likely to be primary, residual, or intrusive (Schiffer, 1987).

Equifinality is a central problem: different processes can create similar material patterns. A concentration of animal bone may represent consumption, craft production, carnivore activity, water sorting, or a mixture of these. Robust interpretation therefore requires diagnostic observations and multiple working hypotheses. Refitting, taphonomy, micromorphology, spatial analysis, use-wear, sediment chemistry, and experimental reference collections can discriminate among alternatives, but rarely eliminate uncertainty entirely.

3.3 Sampling and representativeness

Archaeological populations are usually inaccessible in full. Researchers sample landscapes, sites, deposits, artifacts, and biological materials, and each stage introduces selection. Survey transect spacing affects site discovery; excavation-area placement affects activity reconstruction; mesh size affects the recovery of small bones and artifacts; flotation strategy affects plant remains; and laboratory selection affects chronological and biomolecular conclusions. Because preservation and visibility are also non-random, a large sample may remain biased.

Sampling design should define the target population, sampling frame, inclusion rule, spatial unit, expected variability, and stopping criterion before fieldwork where possible. Stratified, systematic, cluster, adaptive, and judgmental sampling can each be appropriate, but their inferential consequences should be explicit. Destructive analyses require an additional justification: the expected informational gain must outweigh the loss of irreplaceable material and must not disregard community or legal authority.



The sequence is iterative rather than linear; evidence, ethics, and stakeholder priorities may redirect any stage.

Figure 1. Integrated archaeological research lifecycle. The sequence is iterative rather than linear: preservation needs, stakeholder priorities, and unexpected results can redefine the original questions. Original figure by the author.

4. Archaeological Research Workflow

4.1 Research design, permissions, and partnership

A defensible project begins with a question that can be answered by archaeological evidence and a clear statement of why the work is necessary. Research design links theory to observables: it identifies expected material signatures, rival explanations, required scales, methods, sample sizes, analytical pathways, curation plans, and decision points. It should also specify what result would weaken the preferred interpretation. Questions framed only as finding a site, obtaining dates, or generating genomes are method-centered rather than explanation-centered.

Permissions are multi-layered. Legal permits do not automatically constitute ethical legitimacy, and landowner access does not substitute for the authority of descendant, Indigenous, religious, or local communities. Partnership should begin before research questions and sampling plans are fixed. Agreements may cover governance, employment, training, intellectual property, culturally restricted information, publication review, data access, object custody, reburial, benefit sharing, and the conditions under which research should not proceed. Community-based archaeology treats participation as co-production rather than outreach after decisions have been made (Atalay, 2012; Smith & Wobst, 2005).

4.2 Desk-based assessment and predictive reasoning

Before field intervention, archaeologists synthesize maps, previous reports, collections, aerial photographs, satellite data, environmental records, place names, historical documents, oral histories, and heritage inventories. The purpose is not merely to compile background information but to assess previous disturbance, data gaps, sampling bias, access, risk, and alternative explanations. Predictive models can guide survey by estimating where particular site classes or preservation conditions may occur, but they must be validated against independent observations and should not be confused with cultural determinism.

Desk-based work is also a preservation method. Reanalysis of legacy collections and archives can answer new questions without excavation, especially when improved dating, microscopy, or digital methods are available. However, legacy data often contain inconsistent coordinates, undocumented selection, obsolete terminology, and colonial collecting histories. Data cleaning must preserve original values and provenance rather than silently replacing them.

4.3 Survey and prospection

Survey operates from artifact-scale documentation to continental remote sensing. Pedestrian fieldwalking records surface materials and features; systematic shovel testing or augering samples buried deposits; aerial and satellite imagery reveal crop, soil, moisture, shadow, thermal, and spectral contrasts; airborne LiDAR measures topography and can expose subtle relief under vegetation; and geophysical instruments detect contrasts related to magnetism, electrical resistance, conductivity, dielectric properties, or density. The strongest programs combine methods because a feature invisible to one sensor may be clear to another (Conyers, 2013; Doneus, 2013; Kvamme, 2003; Vinci et al., 2025).

Remote sensing is not a substitute for archaeological interpretation or ground validation. Processing choices—filtering, interpolation, terrain modelling, visualization, classification, and thresholding—shape what becomes visible. Sensor anomalies are not self-identifying walls, tombs, or roads; they are measurements requiring contextual evaluation. False positives and false negatives should be documented, and sensitive coordinates should be withheld when disclosure would facilitate looting.

Table 2. Comparative characteristics of common archaeological survey and prospection methods.

Method	Typical strength	Main limitation	Best-use condition	Intervention
Pedestrian survey	Direct observation and artifact characterization	Visibility and observer bias	Exposed or lightly vegetated surfaces with systematic transects	Low
Satellite / aerial imagery	Large-area reconnaissance and diachronic comparison	Resolution, seasonality, atmospheric and vegetation effects	Regional landscape questions and monitoring	None
Airborne LiDAR	Microtopography and canopy-filtered terrain models	Cost, classification error, weak response to flat subsurface features	Wooded or topographically subtle landscapes	None
Drone photogrammetry	Rapid high-resolution orthomosaics and 3D models	Regulation, lighting, vegetation, control-point quality	Site mapping, excavation recording, change detection	None
Magnetometry	Fast mapping of fired, filled, or magnetically contrasting features	Metal interference and geology	Settlement plans, ditches, kilns, pits	None
Electrical resistance	Sensitivity to moisture and structural contrasts	Slower acquisition and soil dependence	Walls, ditches, graves in suitable moisture conditions	None

Method	Typical strength	Main limitation	Best-use condition	Intervention
Ground-penetrating radar	Depth-resolved reflection profiles	Complex interpretation, attenuation in conductive soils	Architecture, graves, voids, stratified targets	None
Coring / augering	Subsurface sequence and paleoenvironmental sampling	Sparse spatial coverage and local disturbance	Deep deposits, wetlands, alluvium, buried landscapes	Low–moderate

4.4 Excavation and recording

Excavation is justified when research, conservation, development impact, or emergency rescue requires information unavailable through less invasive means. Strategies include test excavation, trenches, stepped sections, open-area exposure, block excavation, and micro-excavation in laboratories. The appropriate strategy depends on the research scale and stratigraphic geometry. Arbitrary levels may aid control in homogeneous deposits but should not replace natural or cultural stratigraphy where contexts can be distinguished.

Recording normally integrates context sheets, plans, sections, photographs, coordinates, elevations, samples, finds, conservation notes, and interpretive relationships. Total stations, GNSS, photogrammetry, laser scanning, tablets, and relational databases increase precision and speed, but digital efficiency does not guarantee conceptual consistency. Every field should have a controlled definition, identifiers must remain stable, and edits should be auditable. Daily interpretation is valuable because excavation decisions depend on current understanding, yet provisional interpretations should remain distinguishable from observations (Hodder, 1999; Lucas, 2001).

Recovery procedures should match expected evidence. Fine mesh and flotation improve retrieval of small faunal and botanical remains; column samples preserve vertical variation; bulk samples support sediment, parasite, insect, lipid, and environmental analyses; and uncontaminated sampling protocols are essential for biomolecules. Conservation begins at exposure: metals, organics, pigments, glass, waterlogged materials, and fragile human remains may deteriorate rapidly when temperature, moisture, oxygen, or salts change.

Context, Superposition, and Stratigraphic Reasoning

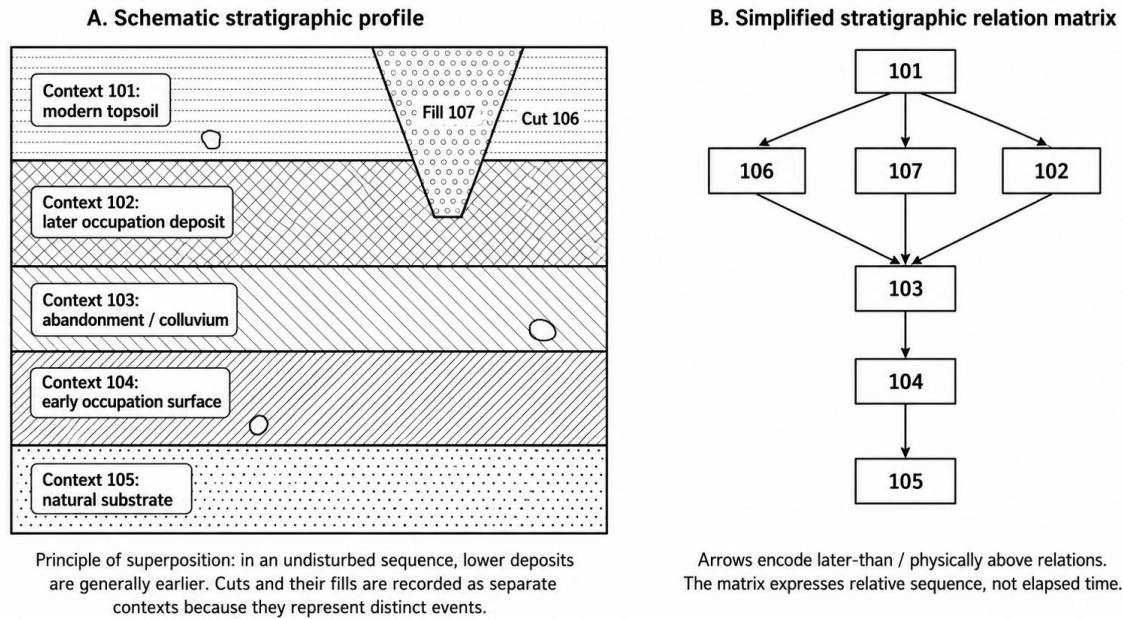


Figure 2. Context, superposition, and stratigraphic reasoning. The profile illustrates why a cut and its fill must be recorded separately from surrounding deposits. The matrix represents relative sequence rather than elapsed time. Original figure by the author.

5. Building Archaeological Chronologies

5.1 Relative chronology

Relative dating orders events without assigning calendar ages. Stratigraphic superposition, cross-cutting relationships, artifact seriation, technological change, architectural sequence, and faunal or floral succession can establish earlier–later relations. Typology is most reliable when categories are explicit, variability is quantified, and dates are independently anchored; otherwise, circularity arises when artifacts date deposits because deposits were used to date artifacts.

Cross-dating transfers chronological information from well-dated assemblages to comparable material, but similarity can result from persistence, revival, trade, or convergent design. Relative chronologies should therefore report uncertainty and regional limits rather than treating named periods as universal time blocks.

5.2 Chronometric methods

Chronometric methods estimate ages using physical, chemical, biological, or historical processes. Radiocarbon dating measures the residual activity or isotopic ratio of carbon-14 in once-living material. The simplified decay relationship is:

$$N(t) = N_0 e^{(-\lambda t)} \quad \text{and} \quad t = (1/\lambda) \ln[N_0/N(t)]$$

where $N(t)$ is the remaining radiocarbon at time t , N_0 is the initial amount, and λ is the decay constant. Archaeological radiocarbon ages require calibration because atmospheric radiocarbon has varied through time. IntCal20 provides the internationally used Northern Hemisphere atmospheric calibration curve to 55,000 calibrated years before present, with separate curves for Southern Hemisphere and marine samples (Reimer et al., 2020). Reservoir effects, old wood, contamination, sample association, and pretreatment can dominate the final uncertainty.

Dendrochronology can provide annual precision where long, regionally cross-dated tree-ring sequences and suitable timbers exist. Luminescence methods estimate the time since minerals were last heated or exposed to light and are useful for ceramics, burned stone, and sediments, but depend on dose-rate reconstruction and adequate resetting. Uranium-series dating is valuable for carbonates and some bones but requires modelling open-system behavior. Potassium–argon and argon–argon methods date volcanic materials and are crucial for deep human prehistory. Archaeomagnetism, paleomagnetism, obsidian hydration, amino-acid racemization, and historical coins or inscriptions may contribute under specific conditions (Aitken, 1998; Renfrew & Bahn, 2020).

Precision is not the same as accuracy. A laboratory result can be analytically precise yet archaeologically irrelevant if the sample is residual, intrusive, poorly associated, or derived from a long-lived material. Sample selection should be based on the event to be dated, such as a short-lived seed from a sealed destruction layer rather than undiagnostic charcoal from a mixed fill.

Table 3. Major relative and chronometric tools used in archaeological chronology.

Method	Materials / event dated	Approximate useful range	Principal strength	Critical cautions
Radiocarbon (^{14}C)	Short-lived plant remains, bone collagen, charcoal, textiles	Recent past to ~50 ka, depending on preservation	Widely applicable; probabilistic calibration	Association, contamination, reservoir effects, old wood, calibration plateaus

Method	Materials / event dated	Approximate useful range	Principal strength	Critical cautions
Dendrochronology	Wood growth rings	Region- and sequence-dependent; often centuries to millennia	Calendar-year precision and environmental data	Requires preserved rings, regional master sequence, outer-ring assessment
OSL / IRSL	Last light exposure of mineral grains in sediment	Commonly hundreds to >100,000 years	Dates sediment deposition directly	Incomplete bleaching, dose-rate heterogeneity, post-depositional mixing
Thermoluminescence	Last heating of ceramics or burned minerals	Typically hundreds to several hundred thousand years	Useful beyond radiocarbon and for heated materials	Dose rate, reheating, sample light exposure
U-series	Carbonates, speleothems, corals; sometimes bones/teeth	~1 ka to >500 ka depending on system	Long range and cave contexts	Open-system uranium movement and detrital correction
K–Ar / ^{40}Ar – ^{39}Ar	Volcanic minerals and eruption events	~10 ka to billions of years	Essential for deep-time volcanic sequences	Reworking, excess argon, relation between eruption and archaeology
Archaeomagnetism	Last heating or deposition recorded by magnetic minerals	Regional secular-variation curve dependent	Can date fired structures and sediments	Regional calibration, disturbance, orientation
Documentary / numismatic	Inscribed objects, coins, archival events	Historically bounded	Potentially exact terminus information	Curated objects, long circulation, copying, calendar conversion

5.3 Bayesian chronological modelling

Bayesian chronology combines calibrated likelihoods with prior information such as stratigraphic order, phase membership, historical constraints, or known duration. In general form, posterior probability is proportional to likelihood multiplied by prior probability:

$$P(\theta | D) \propto P(D | \theta) P(\theta)$$

The method can sharpen estimates for boundaries and event sequences, detect inconsistent dates, and express uncertainty coherently, but it does not rescue poor samples or unjustified priors. Model structure should be archaeologically motivated, alternative models should be compared, outliers handled transparently, and code or model files archived so the chronology can be reproduced (Bronk Ramsey, 2009).

Chronological claims should distinguish laboratory measurement error, calibration uncertainty, model uncertainty, and archaeological uncertainty about association. Reporting only a midpoint or a single calendar range conceals the probability distribution and encourages false precision.

6. Archaeological Science

6.1 Materials characterization, technology, and provenance

Archaeometry applies physical and chemical techniques to archaeological materials. Petrography, scanning electron microscopy, X-ray fluorescence, inductively coupled plasma mass spectrometry, neutron activation analysis, Raman spectroscopy, X-ray diffraction, metallography, and computed tomography can characterize composition, microstructure, manufacturing, alteration, and provenance. Portable instruments support rapid screening but often trade sensitivity, geometry control, or depth resolution for mobility. Analytical results should be validated with standards, blanks, replicates, uncertainty estimates, and independent petrographic or contextual evidence (Pollard et al., 2007).

Provenance is rarely a one-step match between an artifact and a source. Raw materials can be heterogeneous; production can mix sources; recycling can alter composition; and weathering can change surfaces. Statistical classification requires representative source samples and should report overlap and misclassification rather than only assigning a most likely origin. Technological reconstruction likewise integrates composition with chaînes opératoires, experimental archaeology, use-wear, repair, spatial distribution, and social context.

6.2 Isotopes, residues, and environmental proxies

Stable carbon and nitrogen isotopes in bone collagen and other tissues contribute to dietary reconstruction; oxygen and strontium isotopes can inform mobility and geographic histories; sulfur, hydrogen, lead, and other isotope systems address environment, diet, and provenance. Interpretation depends on tissue turnover, local baselines, trophic enrichment, food-web complexity, diagenesis, and the non-uniqueness of isotope signatures. Isotopes identify patterns compatible with particular behaviors, not personal identities or exact places without adequate comparative baselines (Makarewicz & Sealy, 2015).

Organic-residue analysis can detect lipids, proteins, starches, microfossils, smoke, dyes, resins, and absorbed food compounds in pottery, tools, sediments, and dental calculus. Contamination controls and contextual blanks are essential because modern handling, conservation products, soil migration, and laboratory reagents can mimic ancient residues. Archaeobotany, zooarchaeology, palynology, phytoliths, diatoms, insects, parasites, charcoal, and sedimentary DNA reconstruct subsistence and environments, but each proxy has distinct transport, preservation, and taxonomic limits.

6.3 Bioarchaeology, ancient DNA, and proteins

Bioarchaeology investigates human remains in archaeological and social context, drawing on skeletal biology, paleopathology, funerary archaeology, isotopes, histology, and biomolecules. Skeletal indicators are often non-specific: the same lesion can have multiple causes, and absence of a lesion does not imply absence of disease. Age, sex estimation, ancestry-related morphology, trauma, activity, and demography should be reported probabilistically and with population-specific limitations rather than as fixed identities.

Ancient DNA has transformed research on population history, kinship, pathogens, domestication, and species identification. Authentication relies on dedicated clean facilities, negative controls, replication or independent confirmation where appropriate, characteristic molecular damage, fragment length, contamination estimates, and transparent bioinformatics. Sampling of dense skeletal elements can improve DNA yield but may cause disproportionate damage to culturally and scientifically significant remains. Ancient genomics also risks turning complex identities into genetic categories and can affect living communities; interpretation must avoid equating biological ancestry with culture, language, ethnicity, or political belonging (Alpaslan-Roodenberg et al., 2021; Orlando et al., 2021).

Paleoproteomics can identify species and some biological relationships where DNA preservation is poor, while dental calculus and ancient microbiomes extend molecular evidence into diet and health. As with DNA, destructive sampling should be minimal, permissions explicit, destructive locations photographed and measured, remaining material retained, and raw data archived subject to cultural and privacy constraints.

Table 4. Selected archaeological science approaches and safeguards against over-interpretation.

Analytical family	Common outputs	Archaeological applications	Quality requirements	Frequent overreach
Microscopy / petrography	Minerals, inclusions, microstructure, wear	Technology, fabric groups, use, alteration	Representative section, standards, blind checks	Treating visually similar fabrics as identical sources
Elemental / molecular spectroscopy	Elemental or molecular signatures	Composition, pigments, metals, glass, residues	Calibration, matrix correction, blanks, reference materials	Direct provenance assignment from overlapping signatures
Stable isotopes	Isotopic ratios and distributions	Diet, mobility, ecology, provenance	Diagenesis tests, local baselines, tissue knowledge	Assigning one exact birthplace or food from a non-unique value
Ancient DNA	Sequence data, ancestry components, kinship, pathogens	Population history, domestication, species, disease	Clean lab, controls, authentication, contamination estimates	Equating genetic clusters with cultures or ethnic groups
Proteomics	Peptide fingerprints and proteins	Species identification, taxonomy, diet, disease candidates	Damage assessment, blanks, database transparency	Over-specific biological interpretation from sparse peptides

Analytical family	Common outputs	Archaeological applications	Quality requirements	Frequent overreach
Residue and microfossil analysis	Lipids, starch, phytoliths, use residues	Food processing, craft, plant use, vessel function	Contextual controls, contamination audit, multi-proxy confirmation	Assuming detected compound proves exclusive vessel function
Geoarchaeology / micromorphology	Sedimentary fabrics, formation features	Activity surfaces, burning, trampling, depositional process	Undisturbed sampling, spatial control, comparative reference	Reading all microfeatures as intentional human behavior

7. Digital, Landscape, and Computational Archaeology

7.1 GIS, remote sensing, and three-dimensional documentation

Geographic information systems integrate coordinates, terrain, environmental variables, survey observations, site distributions, routes, visibility, and administrative data. GIS supports descriptive mapping, spatial statistics, least-cost and network analysis, viewshed modelling, predictive modelling, and heritage monitoring. Analytical outputs remain sensitive to coordinate accuracy, raster resolution, temporal mismatch, scale, edge effects, and assumptions about movement or perception. Maps should identify uncertainty and avoid displaying restricted locations.

Airborne laser scanning and structure-from-motion photogrammetry have accelerated landscape and site documentation. LiDAR can map subtle relief beneath partial canopy and has become a major tool for landscape archaeology, while drones can produce centimeter-scale orthomosaics and models for excavation, architecture, and change detection (Campana, 2017; Doneus, 2013; Vinci et al., 2025). Yet no visualization is neutral: hillshade direction, vertical exaggeration, filtering, openness, and color ramps influence feature recognition. Original point clouds, processing parameters, ground-control information, and classification accuracy should be retained.

Three-dimensional models can preserve geometry, enable remote measurement, support conservation, and communicate sites. They are not substitutes for physical material or contextual records, and visually realistic surfaces may conceal interpolation and missing data. Scale bars, coordinate systems, texture provenance, decimation history, error estimates, and licensing should accompany models.

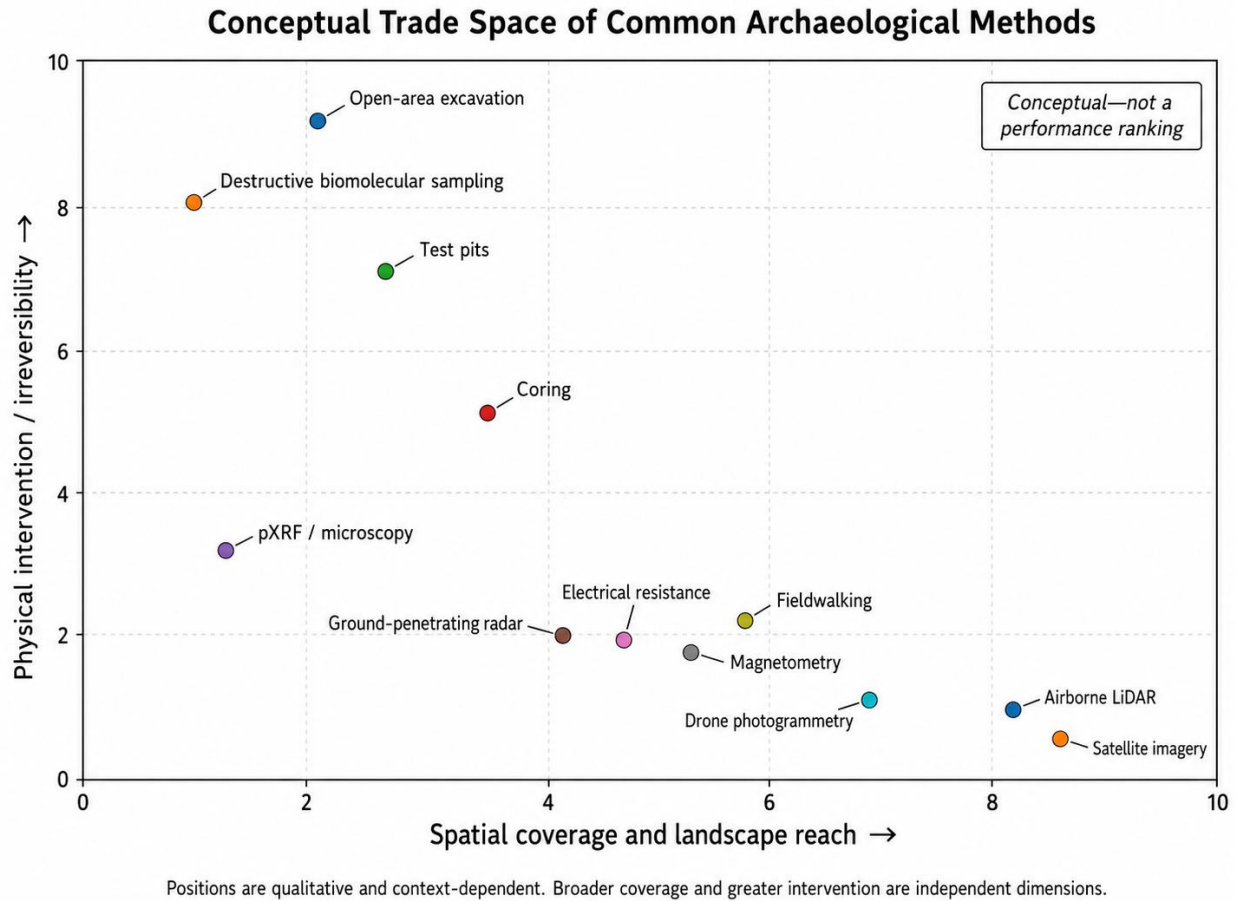


Figure 3. Conceptual trade space of common archaeological methods. Positions are qualitative and context-dependent; the figure emphasizes that broader coverage and greater intervention are independent dimensions and that no method dominates every research problem. Original figure by the author.

7.2 Artificial intelligence and computational modelling

Machine learning can classify remote-sensing features, cluster compositional data, recognize ceramic or lithic forms, transcribe texts, reconstruct fragments, and identify patterns in large archives. Simulation and agent-based modelling can explore how local rules generate settlement, exchange, demographic, or land-use patterns. These methods are most useful when they formalize assumptions and produce testable predictions rather than merely improving visual fit.

Major risks include biased training data, spatial leakage between training and test sets, class imbalance, undocumented preprocessing, opaque feature selection, and the seduction of high accuracy from non-independent samples. Archaeological classes are themselves interpretations and may embed colonial typologies or uneven research

histories. Models should be evaluated on independent regions or time periods, report precision and recall by class, retain expert review, and document failures. Generative AI can assist drafting, coding, image segmentation, or metadata work, but it can also fabricate references, erase uncertainty, and reproduce inequity. Human accountability for claims, permissions, and data remains non-delegable.

7.3 FAIR data, reproducibility, and digital preservation

Archaeological excavation is destructive, which creates a strong obligation to preserve the record of the investigation. FAIR principles—findable, accessible, interoperable, and reusable—encourage persistent identifiers, rich metadata, standardized vocabularies, accessible protocols, and clear reuse conditions (Wilkinson et al., 2016). The Archaeology Data Service applies these principles to long-term archaeological archiving and emphasizes sustainable access and reuse (Archaeology Data Service, n.d.).

FAIR does not mean that all data must be openly released. Human remains, sacred knowledge, personal information, precise site coordinates, and information governed by communities may require restricted, tiered, delayed, or non-public access. Responsible data management balances reproducibility with the risk of looting, surveillance, stigmatization, and loss of cultural authority. A data management plan should address file formats, metadata, version control, identifiers, licences, repository selection, embargoes, backups, software dependencies, and the survival of relational links between contexts, samples, images, objects, and analyses.

8. Interpretation, Explanation, and Theory

8.1 From observations to claims

Archaeological reasoning moves through several levels: observation, classification, pattern description, inference about process, and broader explanation. Problems arise when these levels are collapsed—for example, when a spatial cluster is labelled a household before alternative formation processes are considered, or when a typological unit is treated as an ethnic group. Good reporting distinguishes measured properties from analytical categories and interpretations.

Convergent evidence is stronger than repeated use of the same proxy. If chronology, architecture, botanical remains, use-wear, residues, and spatial organization independently support a food-processing interpretation, confidence increases. By contrast, multiple measurements derived from one disturbed deposit are not truly independent. Claims

should specify their scale: evidence for one burial, household, settlement, region, or century should not automatically be generalized to an entire culture or population.

8.2 Theoretical pluralism and reflexivity

Theory identifies which questions are important and how material patterns connect to human action. Ecological and evolutionary approaches may model adaptation, transmission, demography, and niche construction; practice and agency approaches examine how repeated actions reproduce or transform social structures; political-economy and world-systems approaches study labor, inequality, and exchange; phenomenological and landscape approaches consider embodied experience; network analysis explores relations among people, objects, and places; and materiality approaches examine how things participate in social life. These perspectives can generate complementary or competing expectations.

Reflexivity recognizes that archaeological knowledge is produced by institutions, funding priorities, field hierarchies, legal regimes, and the identities and interests of researchers. Field diaries, transparent interpretive changes, collaborative authorship, and documentation of stakeholder disagreement can make this production visible. Reflexivity does not imply that all interpretations are equally valid; rather, it strengthens evaluation by exposing assumptions and the social conditions under which evidence was created (Hodder, 1999).

8.3 Uncertainty and causal explanation

Uncertainty should be represented at each inferential step. Spatial error, measurement error, chronological probability, classification uncertainty, taphonomic ambiguity, sampling bias, and model dependence are different quantities and should not be merged into a vague caveat. Sensitivity analysis can show whether conclusions change under alternative chronologies, classifications, source baselines, or spatial parameters.

Causal explanations require mechanisms. Correlation between climate and settlement change is insufficient unless the proposed pathway—such as crop failure, water availability, conflict, migration, or institutional adaptation—is supported by appropriate temporal resolution and independent evidence. Human societies respond variably to environmental and political pressures; archaeological narratives should avoid monocausal collapse stories when continuity, adaptation, inequality, and regional diversity are visible.

9. Ethics, Law, and Community Governance

9.1 Stewardship and the finite archaeological record

Stewardship treats archaeological sites, collections, records, and knowledge as a finite inheritance requiring long-term care. The Society for American Archaeology's current ethics framework emphasizes collaborative management, public benefit, accountability, professional conduct, and preservation. Because excavation consumes context, research should prefer existing collections and non-invasive methods when they can answer the question, excavate only what can be documented and curated, and budget for conservation and archive deposition from the outset (Society for American Archaeology, 2024).

The antiquities market can sever objects from context and incentivize clandestine excavation. The UNESCO 1970 Convention establishes international principles for preventing illicit import, export, and transfer of ownership of cultural property. Researchers, journals, museums, and collectors should conduct provenance due diligence, avoid publishing or authenticating recently surfaced unprovenanced objects in ways that increase market value, and cooperate with lawful return and restitution processes (UNESCO, 1970).

9.2 Indigenous, descendant, and local community authority

Archaeological materials may be scientific evidence, ancestors, sacred entities, land claims, or sources of community continuity. These meanings are not mutually reducible. The World Archaeological Congress code requires respect for Indigenous peoples and affected communities, including permission, culturally appropriate practice, and recognition of community interests in human remains and sacred objects (World Archaeological Congress, n.d.). Engagement should therefore affect research design, not merely communication strategy.

Power-sharing may include community-controlled research questions, paid roles, training, co-authorship, language access, local repositories, consent for images, control of restricted knowledge, review before publication, and the right to decline research. Agreement is not always possible, and communities are not internally uniform. Projects should document who was consulted, how representatives were selected, what disagreements remained, and how decisions were made without treating consultation as a box-checking exercise.

9.3 Human remains and destructive biomolecular research

Human remains require dignity, legal compliance, and attention to descendant and cultural relationships. Scientific value does not automatically override funerary, religious, or political concerns. Internationally proposed ancient-DNA guidelines emphasize legal compliance, detailed planning, minimum damage, data availability for scrutiny, and engagement with stakeholders from the beginning of research (Alpaslan-Roodenberg et al., 2021). These are minimum principles rather than substitutes for local governance.

Destructive sampling proposals should identify the exact question, demonstrate that non-destructive alternatives are inadequate, specify expected sample size and success probability, document previous sampling, plan conservation of remaining material, and state how results and residual samples will be returned or governed. Repeated sampling by separate projects can cumulatively destroy rare elements; collection-level sampling registers are therefore essential.

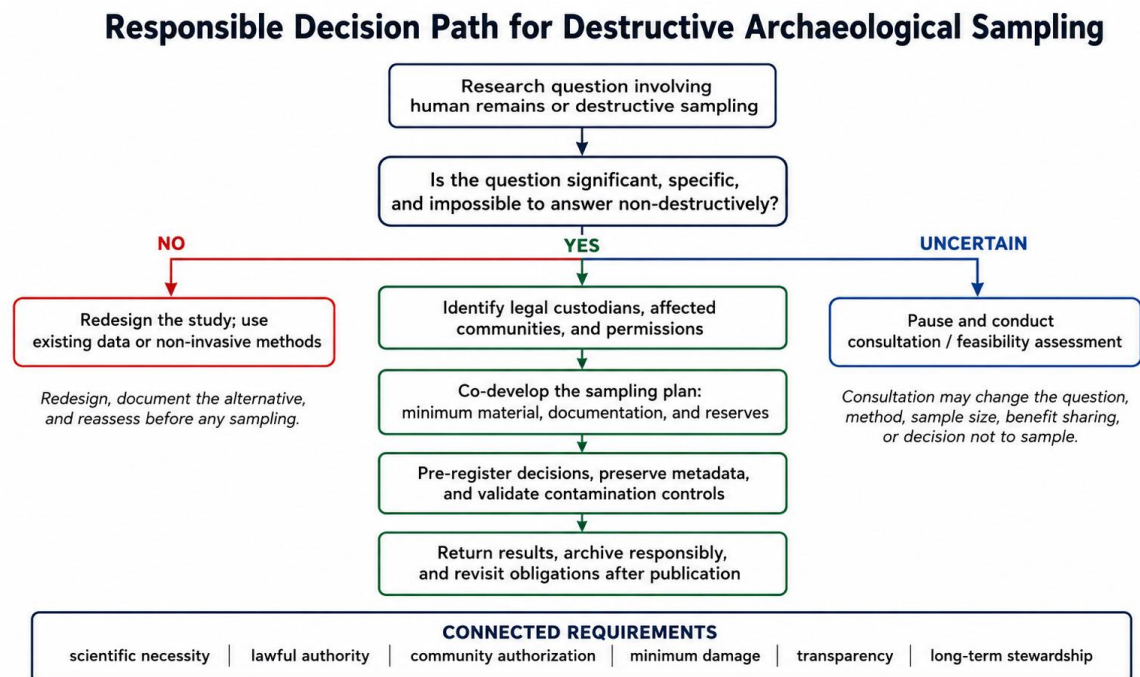


Figure 4. Responsible decision path for destructive archaeological sampling. Legal permission, community authorization, scientific justification, and minimum damage are treated as connected requirements. Original figure by the author.

9.4 Publication, authorship, and public communication

Authorship should reflect substantial intellectual contributions rather than status, funding control, or access to collections. Local archaeologists, conservators, technicians,

community researchers, and data specialists should be credited appropriately. Methods and negative results should be reported sufficiently to prevent unnecessary repeat sampling. Conflicts of interest include commercial valuation, private collections, development contracts, political sponsorship, and media arrangements, not only direct financial payments.

Public archaeology is necessary because archaeological claims influence identity, heritage policy, tourism, and conflict. Communication should distinguish evidence from speculation, avoid sensational claims of lost civilizations or biological purity, and explain uncertainty without making the research inaccessible. Site publicity must be balanced against looting and visitor damage. When misinformation spreads, archaeologists should correct it with transparent evidence and acknowledge legitimate uncertainty rather than relying on professional authority alone.

Table 5. Ethical and governance checklist for archaeological research.

Decision area	Minimum question	Required record	Reason to pause or stop
Site intervention	Can the question be answered from archives, collections, survey, or monitoring?	Alternatives assessment and excavation justification	No clear research or conservation benefit
Community governance	Who holds legal, cultural, ancestral, or land-based authority?	Consultation process, agreements, unresolved objections	Affected authority refuses permission or representation is inadequate
Human remains	Is treatment dignified, legal, and culturally appropriate?	Custody, permissions, handling and disposition plan	Uncertain legality, contested custody, or unacceptable harm
Destructive sampling	Is the sample minimal and expected information gain proportionate?	Sampling map, amount removed, success criteria, residual material	Redundant sampling, weak association, no curation plan
Provenance and trade	Is lawful and ethical collecting history documented?	Ownership chain, export/import records, due diligence	Recent unprovenanced market appearance or signs of illicit excavation
Data release	Would openness expose sites, people, sacred knowledge, or community-controlled information?	Access classification, licence, embargo, coordinate policy	Material risk cannot be mitigated by tiered access

Decision area	Minimum question	Required record	Reason to pause or stop
Authorship and benefit	Have contributors and communities received fair credit and benefit?	Contribution statement, compensation, dissemination plan	Extractive publication or exclusion of substantive contributors

10. Archaeology and Contemporary Challenges

10.1 Climate change, conflict, development, and looting

Archaeological heritage is threatened by sea-level rise, erosion, wildfire, thawing permafrost, desertification, intensified storms, agriculture, mining, urban growth, infrastructure, armed conflict, and looting. These processes operate faster than traditional project cycles. Risk-based survey, remote monitoring, community site stewardship, emergency documentation, selective excavation, and digital archiving can reduce loss, but triage requires transparent criteria because not everything can be preserved in place.

Climate change also alters interpretation. Newly exposed materials may appear exceptionally preserved but deteriorate rapidly; changing vegetation affects remote-sensing visibility; and modern environmental analogies may no longer match past conditions. Archaeological long-term records can contribute to resilience research by documenting previous adaptation, land management, and inequality, but they should not be reduced to simple lessons detached from historical institutions and power.

10.2 Cultural-resource management and rescue archaeology

A large proportion of archaeological work occurs in advance of development. Cultural-resource management and rescue archaeology can generate extensive regional datasets and protect sites that would otherwise be destroyed. Tension arises when commercial schedules constrain research, reporting, curation, or public access. Procurement systems should evaluate quality, independence, worker safety, archive completion, and dissemination rather than lowest price alone.

Mitigation hierarchies should prefer avoidance, redesign, and preservation in situ before excavation. When excavation is necessary, the project should create a proportionate research design, integrate specialist sampling early, and fund post-excavation analysis and deposition. Unpublished backlogs represent an ethical failure because destructive fieldwork has occurred without converting the removed record into accessible knowledge.

10.3 Grand challenges and interdisciplinary contribution

Archaeology contributes to major questions about migration, urbanization, inequality, food systems, disease, technological change, human–environment interaction, colonization, resilience, and the long-term consequences of institutions. Grand-challenge research is strongest when large comparative datasets preserve local context and when global models do not erase regions with limited research infrastructure (Kintigh et al., 2014).

Interdisciplinary collaboration should be symmetric. Archaeologists contribute formation-process knowledge, chronology, context, and social interpretation; laboratory scientists contribute measurement and validation; historians and linguists contribute documentary and language evidence; geoscientists contribute landscape processes; and communities contribute knowledge, governance, and priorities that may not exist in academic archives. Collaboration fails when one discipline treats another only as a service provider.

11. An Integrated Framework for High-Quality Archaeological Research

The synthesis supports a practical framework based on twelve linked commitments. These commitments are applicable to field projects, collections research, laboratory studies, digital projects, and heritage interventions.

Question before technique. Define the historical or anthropological problem, expected evidence, rival explanations, and decision criteria before selecting instruments.

Partnership before extraction. Identify rights-holders and co-design governance, benefit, access, and disposition arrangements before permits and sampling are finalized.

Context before object. Prioritize provenience, association, stratigraphy, and formation processes; isolated objects support narrower claims.

Least intervention. Use archives, collections, survey, and non-invasive methods where they can answer the question; justify irreversible work.

Sampling transparency. Specify target populations, recovery procedures, selection rules, missingness, and the consequences of bias.

Chronological realism. Date events rather than convenient materials, calibrate appropriately, model sequence probabilistically, and report full uncertainty.

Independent convergence. Seek agreement among evidence types with different error structures rather than multiplying correlated measurements.

Analytical validation. Use standards, controls, replicates, contamination audits, sensitivity analysis, and documented software.

Interpretive restraint. Separate observations from categories and explanations; avoid equating artifacts, genes, languages, and identities.

Reproducible but responsible data. Archive metadata, code, models, and raw outputs while restricting sensitive coordinates and culturally governed knowledge.

Curation as a research cost. Budget conservation, collection storage, digital preservation, and publication as core components, not optional aftercare.

Accountable communication. Credit contributors, disclose limitations and assistance, return results, and correct sensational or harmful interpretations.

12. Limitations of This Review

The paper is intentionally broad and cannot replace specialist protocols for excavation safety, conservation, osteology, geophysics, radiocarbon pretreatment, isotope preparation, ancient DNA, statistics, or national heritage law. Method performance is context-specific, and the qualitative comparisons in the tables should be converted into locally validated protocols before field or laboratory use.

The literature of archaeology is multilingual and regionally uneven. English-language publications and globally visible organizations are overrepresented in this synthesis, while many Indigenous, local, contract-archaeology, museum, and government reports remain difficult to discover. The review also does not attempt a comprehensive historiography of archaeological theory. Future editions should be developed with regional co-authors and a formal evidence-mapping protocol.

Finally, a publishable review requires author-led critical verification. All citations, journal requirements, legal statements, and local ethical procedures should be checked against the target submission context. No generic framework can determine whether a particular excavation or analysis is legitimate without site-specific evidence and consultation.

13. Conclusion

Archaeology is a disciplined reconstruction of human histories from incomplete and transformed material traces. Its distinctive strength lies in linking objects, bodies, buildings, sediments, landscapes, documents, and memories across scales of time and space. Its distinctive vulnerability is that the process of producing knowledge can consume the evidence, detach it from communities, or convert uncertainty into authoritative stories.

A rigorous archaeology therefore joins scientific validation with humanistic interpretation and ethical governance. Stratigraphy and formation processes establish what observations mean; sampling and chronology define the limits of generalization; archaeological science expands the evidence; theory connects patterns to human action; and community partnership determines whether research is legitimate and beneficial. Digital tools, ancient biomolecules, LiDAR, and artificial intelligence will continue to transform the field, but none can compensate for weak context, unrepresentative samples, opaque models, or extractive relationships.

The future of archaeology should be measured not by the amount excavated or the novelty of instruments, but by the durability, accountability, and explanatory value of the knowledge created. The best projects ask consequential questions, intervene minimally, preserve uncertainty, share authority, archive responsibly, and leave both archaeological records and research relationships stronger than they found them.

Declarations

Author: SAMUELSON G.

Author contribution: Conceptualization, review design, synthesis, critical interpretation, and manuscript approval: SAMUELSON G.

Funding: No external funding was received for this review manuscript.

Conflicts of interest: The author declares no financial or non-financial conflicts of interest relevant to this manuscript.

Ethics approval: Not applicable. This review did not involve living participants, excavation, intervention, or analysis of human or animal remains.

Consent to participate: Not applicable.

Consent for publication: The author approves publication of the manuscript.

Data availability: No original research dataset was generated. The conceptual figures are original to this manuscript; source literature is listed in the references.

Acknowledgements: The author acknowledges the scholars, heritage professionals, communities, and institutions whose published methods and ethical frameworks inform this review. No individual endorsement is implied.

References

1. Aitken, M. J. (1998). *An Introduction to Optical Dating: The Dating of Quaternary Sediments by the Use of Photon-Stimulated Luminescence*. Oxford University Press.
2. Alpaslan-Roodenberg, S., Anthony, D., Babiker, H., et al. (2021). Ethics of DNA research on human remains: Five globally applicable guidelines. *Nature*, 599, 41–46. <https://doi.org/10.1038/s41586-021-04008-x>
3. Archaeology Data Service. (n.d.). FAIR data. University of York.
4. Atalay, S. (2012). *Community-Based Archaeology: Research with, by, and for Indigenous and Local Communities*. University of California Press.
5. Binford, L. R. (1962). Archaeology as anthropology. *American Antiquity*, 28(2), 217–225. <https://doi.org/10.2307/278380>
6. Bronk Ramsey, C. (2009). Bayesian analysis of radiocarbon dates. *Radiocarbon*, 51(1), 337–360. <https://doi.org/10.1017/S0033822200033865>
7. Brown, T. A., & Brown, K. A. (2011). *Biomolecular Archaeology: An Introduction*. Wiley-Blackwell.
8. Campana, S. (2017). Drones in archaeology: State-of-the-art and future perspectives. *Archaeological Prospection*, 24(4), 275–296. <https://doi.org/10.1002/arp.1569>
9. Conyers, L. B. (2013). *Ground-Penetrating Radar for Archaeology* (3rd ed.). AltaMira Press.
10. DeNiro, M. J. (1985). Postmortem preservation and alteration of in vivo bone collagen isotope ratios in relation to palaeodietary reconstruction. *Nature*, 317, 806–809. <https://doi.org/10.1038/317806a0>
11. Doneus, M. (2013). Openness as visualization technique for interpretative mapping of airborne lidar derived digital terrain models. *Remote Sensing*, 5(12), 6427–6442. <https://doi.org/10.3390/rs5126427>
12. Harris, E. C. (1989). *Principles of Archaeological Stratigraphy* (2nd ed.). Academic Press.
13. Hodder, I. (1999). *The Archaeological Process: An Introduction*. Blackwell.
14. Killick, D. (2015). The awkward adolescence of archaeological science. *Journal of Archaeological Science*, 56, 242–247.
15. Kintigh, K. W., Altschul, J. H., Beaudry, M. C., et al. (2014). Grand challenges for archaeology. *American Antiquity*, 79(1), 5–24. <https://doi.org/10.7183/0002-7316.79.1.5>
16. Kvamme, K. L. (2003). Geophysical surveys as landscape archaeology. *American Antiquity*, 68(3), 435–457. <https://doi.org/10.2307/3557103>
17. Lipe, W. D. (1974). A conservation model for American archaeology. *Kiva*, 39(3–4), 213–245.

18. Lucas, G. (2001). *Critical Approaches to Fieldwork: Contemporary and Historical Archaeological Practice*. Routledge.
19. Makarewicz, C. A., & Sealy, J. (2015). Dietary reconstruction, mobility, and the analysis of ancient skeletal tissues: Expanding the prospects of stable isotope research in archaeology. *Journal of Archaeological Science*, 56, 146–158. <https://doi.org/10.1016/j.jas.2015.02.035>
20. Orlando, L., Allaby, R., Skoglund, P., et al. (2021). Ancient DNA analysis. *Nature Reviews Methods Primers*, 1, Article 14.
21. Orton, C., & Hughes, M. (2013). *Pottery in Archaeology* (2nd ed.). Cambridge University Press.
22. Pollard, A. M., Batt, C. M., Stern, B., & Young, S. M. M. (2007). *Analytical Chemistry in Archaeology*. Cambridge University Press.
23. Reimer, P. J., Austin, W. E. N., Bard, E., et al. (2020). The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0–55 cal kBP). *Radiocarbon*, 62(4), 725–757. <https://doi.org/10.1017/RDC.2020.41>
24. Renfrew, C., & Bahn, P. (2020). *Archaeology: Theories, Methods and Practice* (8th ed.). Thames & Hudson.
25. Schiffer, M. B. (1987). *Formation Processes of the Archaeological Record*. University of New Mexico Press.
26. Society for American Archaeology. (2024). *Principles of Archaeological Ethics*. Adopted 7 March 2024.
27. Society for American Archaeology. (n.d.). What is archaeology?.
28. Smith, C., & Wobst, H. M. (Eds.). (2005). *Indigenous Archaeologies: Decolonizing Theory and Practice*. Routledge.
29. Trigger, B. G. (2006). *A History of Archaeological Thought* (2nd ed.). Cambridge University Press.
30. UNESCO. (1970). *Convention on the Means of Prohibiting and Preventing the Illicit Import, Export and Transfer of Ownership of Cultural Property*. Paris: United Nations Educational, Scientific and Cultural Organization.
31. Vinci, G., Vanzani, F., Fontana, A., & Campana, S. (2025). LiDAR applications in archaeology: A systematic review. *Archaeological Prospection*, 32(1), 81–101. <https://doi.org/10.1002/arp.1931>
32. Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>
33. Wiseman, J., & El-Baz, F. (Eds.). (2007). *Remote Sensing in Archaeology*. Springer.
34. World Archaeological Congress. (n.d.). Code of Ethics.